1ª Week: Meeting 22/06/2026

Gabriel de Carvalho 15/06/2026, 09:27

**TAREFA**

1- Escolha dos termos usados na busca
2- Busca realizada em quais buscadores (ex. Google Scholar)
3- Seleção dos artigos a serem lidos ( e quais critérios foram usados )
4- Resumo dos selecionados

Como ler um artigo científico:

1- Título (Saber se é interessante)
2- Resumo/Abstract (Saber se é interessante)
3- Conclusão (Entender o que foi descoberto)
4- Introdução (Entender melhor o contexto) (editado)

**Modelo de Trabalho do Laboratório**

Cada aluno é responsável por planejar sua própria semana.

**Fluxo semanal:**

1- O aluno cria uma Issue no GitHub. Na Issue descreve:
a- O que pretende realizar;
b- Quais entregas pretende produzir.

2- O orientador analisa a proposta. A Issue pode ser:
a- Aprovada;
b- Alterada;
c- Rejeitada com sugestões.

3- Ao final da semana o aluno registra os resultados obtidos. (editado)

**Search performed on the search engine:**

Google Scholar

**Search term:**

Intelligent Adaptive Games

**Articles:**

**Title:** Filling the Need for Intelligent, Adaptive Non-Player Characters

**Link:**

https://www.researchgate.net/profile/Douglas-Maxwell-5/publication/292984684_Filling_the_Need_for_Intelligent_Adaptive_Non-Player_Characters/links/56b3fb6008ae5deb2657e80b/Filling-the-Need-for-Intelligent-Adaptive-Non-Player-Characters.pdf

**Title**:

To Create Intelligent Adaptive Game Opponent by Using Monte-Carlo for the Game of Pac-Man

**Link:**

https://ieeexplore.ieee.org/abstract/document/5366867

2ª Week: Meeting 29/06/2026

**1º ARTICLE:**

**Search performed on the search engine:**

Google Scholar

**Search term:**

Intelligent Adaptive Games

**Article Title:**

To Create Intelligent Adaptive Game Opponent by Using Monte- Carlo for the Game of Pac-Man

**Link:**

https://ieeexplore.ieee.org/abstract/document/5366867

**Resume:**

**Prompt:** Summarize this so a 10-year-old can understand it.

Imagine you're playing **Pac-Man**, but instead of the ghosts always following the same rules, they can **think and get smarter**.

Here's what the paper is about in kid-friendly language:

- The researchers wanted to make the ghosts in Pac-Man more fun to play against.

- They used a method called **Monte Carlo**, which is like having the ghosts **imagine lots of possible moves very quickly** before choosing the best one.
- ⚡ If the ghosts have **more time to think**, they usually make better choices and become harder to beat. If they have less time, they're easier to beat. This lets the game automatically become easier or harder.
- The problem is that all this thinking takes a lot of computer power and can slow the game down.
- So the researchers trained an **artificial neural network** (a simple kind of AI that learns from examples) by watching the Monte Carlo method play. After learning, the new AI could make good decisions **much faster**.
- In the end, they found that the trained AI played almost as well as the Monte Carlo method but without making the game slow, so it could work better in real games.

**In one sentence:**

**The paper teaches Pac-Man's ghosts to become smarter by practicing lots of pretend games, and then helps them remember what they learned so they can play well without thinking for a long time.**

**ChatGPT prompt:** Summarize the following text in 300 words.

This paper explores how to create a more intelligent and adaptive opponent AI for the game Pac-Man using the Monte Carlo Tree Search (MCT) algorithm. Traditional game opponents often rely on fixed, scripted rules that make their behavior predictable and difficult to adjust. In contrast, Monte Carlo methods enable the AI to evaluate many possible future moves through repeated simulations, allowing it to choose actions that are more likely to succeed. This approach creates opponents that can dynamically adapt their level of difficulty, making gameplay more engaging.

To test this idea, the researchers modified the original Pac-Man game by simplifying some of its rules. Two ghosts cooperate to chase Pac-Man, while Monte Carlo simulations help determine their best movement at each step. The study found that the AI's performance improves as it is given more computation time because additional simulations produce better decisions. However, after approximately 400 milliseconds of computation, further improvements become minimal while causing delays that players would find unacceptable.

Because Monte Carlo simulations require significant computing power, they are not ideal for real-time or online games. To address this limitation, the researchers trained an Artificial Neural Network (ANN) using gameplay data generated by the Monte Carlo algorithm. The neural network learned to imitate the high-quality decisions produced by Monte Carlo while requiring far less computation during gameplay. Inputs to the network included the positions of Pac-Man and the ghosts, distances between them, nearby walls, and other game-state information. The trained model successfully controlled the ghosts with performance comparable to the Monte Carlo approach but with much greater speed and stability.

The authors conclude that combining Monte Carlo simulations with neural network learning provides an effective way to build adaptive game AI. Their approach enables adjustable game difficulty, maintains strong opponent performance, and reduces computational cost, making it suitable for real-time games and potentially applicable to many other game genres beyond Pac-Man.

**2º ARTICLE:**

**Origin:**

Reference of article 1

**Article Tittle:**

powerGhosts & defensiveGhosts – Enhanced Ghost Team Controller based on Ant Colony Optimization for Ms. Pac-Man

**Link:**

https://ieeexplore.ieee.org/document/9654859

**3º ARTICLE:**

**Origin:**

Reference of article 2

**Article Tittle:**

fairGhosts – Ant Colony Controlled Ghosts for Ms. Pac-Man

**Link:**

https://ieeexplore.ieee.org/document/7744325

**Resume:**

**ChatGPT prompt:** Summarize the following text in 300 words.

This paper presents fairGhosts, an improved ghost controller for the game Ms. Pac-Man based on Ant Colony Optimization (ACO), a swarm intelligence algorithm inspired by how real ants find efficient paths to food. The goal is to create ghost agents that cooperate effectively to minimize Ms. Pac-Man's score while operating within the strict 40-millisecond decision limit imposed by the Ms. Pac-Man vs. Ghosts competition.

The proposed controller models the game maze as a graph and employs two types of artificial ants. ExplorerAnts investigate the maze and leave pheromone trails representing promising paths, while HunterAnts use this information to determine the best routes for the ghosts to pursue or evade Ms. Pac-Man, depending on whether they are in their normal or edible state. To encourage teamwork, ghosts reduce pheromone levels on their chosen paths so that other ghosts are more likely to select alternative routes, increasing the chances of trapping Ms. Pac-Man instead of following each other.

Compared with the original ACO framework, the authors simplified the algorithm, corrected the heuristic guiding ghost movement, removed unnecessary features, and reduced the number of configurable parameters from ten to seven. They then optimized these parameters using a genetic algorithm, which evaluated many parameter combinations over repeated game simulations to identify those producing the strongest ghost performance.

To evaluate fairGhosts, the controller was tested against several leading Ms. Pac-Man and ghost controllers from the 2012 competition. Although a direct comparison with the original implementation was impossible due to changes in the simulator, fairGhosts consistently achieved competitive results. Its overall performance placed it within the upper third of all ghost controllers, with realistic potential to rank among the top ten.

The authors conclude that Ant Colony Optimization is an effective approach for developing cooperative, adaptive game AI. The fairGhosts controller demonstrates that swarm intelligence can produce strong real-time decision making in dynamic game environments while remaining computationally efficient and flexible enough for future improvements and applications beyond Ms. Pac-Man.

## 4º ARTICLE:

**Origin:**

Reference of article 1

**Article Tittle:**

*HearthBot*: An Autonomous Agent Based on Fuzzy ART Adaptive Neural Networks for the Digital Collectible Card Game *HearthStone*

**Link:**

https://ieeexplore.ieee.org/document/8015141

**Resume:**

**ChatGPT prompt:** Summarize the following text in 300 words.

This paper presents HearthBot, an adaptive artificial intelligence (AI) agent designed to play the digital card game Hearthstone more effectively than traditional search-based methods. Hearthstone is a highly complex game with imperfect information, hidden cards, and an enormous number of possible game states, making it impractical to create optimal strategies manually or through exhaustive search. The authors estimate that the game contains approximately $2.85 \times 10^{51}$ possible states, demonstrating the need for advanced machine learning techniques rather than conventional algorithms.

To address this challenge, the researchers model the game as a Partially Observable Markov Decision Process (POMDP) and represent game environments using key features such as player health, cards in hand, minions on the board, and battlefield strength. Actions are encoded compactly based on card interactions and event categories, allowing the AI to efficiently process complex game situations.

The main contribution of the paper is an improved Adaptive Resonance Associative Map (ARAM) neural network. Instead of relying on traditional similarity measures that often overgeneralize different game situations, the authors introduce a normalized Manhattan distance metric to compare game states more accurately. This modification improves learning, prediction, and categorization while reducing data corruption and increasing the precision of decision making.

HearthBot follows a three-step process: it senses the current game state, predicts the reward of every possible action using the neural network, and selects the action with the highest predicted reward. During training, it learns by comparing simulated outcomes before and after each action, allowing it to refine its decision-making over time. The system was implemented using GPU acceleration to efficiently process millions of neural representations.

The researchers evaluated HearthBot against the Monte Carlo Tree Search (MCTS) agent provided by the Metastone simulator using multiple competitive Hearthstone decks. Experimental results showed that HearthBot consistently outperformed MCTS, achieving approximately 30% higher average win rates against familiar decks and around 21% higher win rates against previously unseen decks. The improved neural network also demonstrated strong generalization while maintaining efficient memory usage and scalable learning.

The authors conclude that combining an improved adaptive neural network with efficient state representation provides a powerful solution for decision making in complex, imperfect-information games. HearthBot demonstrates that adaptive learning can outperform traditional search methods while remaining capable of transferring learned knowledge to new situations, making the approach promising for future game AI research and other real-time decision-making applications.